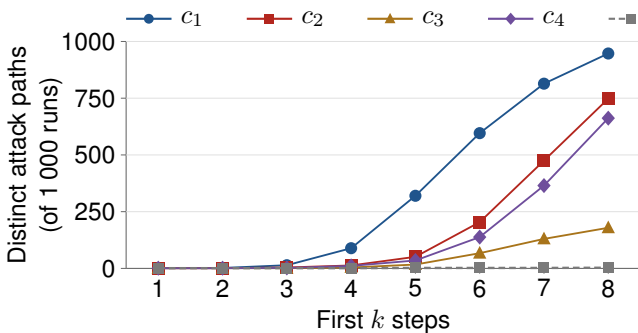


**(a)** Relative tactic occurrence (%)

|                  |                      | APT attacker model |       |       |       | baseline attacker |
|------------------|----------------------|--------------------|-------|-------|-------|-------------------|
|                  |                      | $c_1$              | $c_2$ | $c_3$ | $c_4$ |                   |
| scan host        | Reconnaissance       | 5                  | 1     | 1     | 2     | 2                 |
| enumerate host   | Lateral movement     | 9                  | 8     | 6     | 14    | 25                |
| scan port        | Discovery            | 8                  | 12    | 17    | 9     | 25                |
| exploit          | Initial access       | 3                  | 6     | 1     | <1    | 23                |
|                  | Execution            | 8                  | 14    | 9     | 17    |                   |
|                  | Privilege escalation | 3                  | 3     | 7     | 3     |                   |
| brute force      | Credential access    | 10                 | 3     | 1     | 10    | 16                |
| scan neighbours  | Command and control  | 17                 | 14    | 11    | 19    | 9                 |
| <i>dwel-only</i> | Resource development | 0                  | 0     | 1     | 3     |                   |
|                  | Persistence          | 8                  | 13    | 10    | 4     |                   |
|                  | Stealth              | 14                 | 16    | 16    | 10    |                   |
|                  | Defense impairment   | 1                  |       |       | 5     |                   |
|                  | Collection           | 7                  | <1    | 4     | 4     |                   |
|                  | Exfiltration         | 6                  |       | 4     |       |                   |
|                  | Impact               | <1                 | 8     | 12    |       |                   |

**(b)** Distinct attack paths**(c)** Attack confidentiality